

Blockchain-Based Traffic Accident Handling Protocol Without Third Party for VANETs

Qi Xie¹, Zixuan Ding¹, Qingyun Xie¹, Xiao Tan¹, Debiao He², *Member, IEEE*, and Wen Tang

Abstract—In vehicular ad-hoc networks (VANETs), existing traffic accident handling schemes are adoptable only in the scenario with roadside unit (RSU) deployment, and the generation of accident reports relies on RSUs and witness vehicles. Without the confirmation from involved vehicles in the accident, it will probably cause disputes afterward, and RSU captured attacks may also affect authentication of vehicles and correctness of accident reports. To address the above issues, we propose a vehicle to vehicle (V2V) and vehicle to RSU (V2R) authentication and traffic accident handling protocol, in which accident vehicles are sufficient to generate accident reports, thereby enhancing autonomy of the vehicular communication system and reducing reliance on external infrastructure. Furthermore, for ensuring integrity and traceability of accident reports, we utilize blockchain to keep registration information and jointly signed reports, which realizes efficient and secure mutual authentications in V2V and V2R protocols. Finally, for preserving privacy of vehicles, we integrate elliptic curve cryptosystem (ECC) and symmetric encryption to design a dynamic pseudo-identity strategy, which still allows the registration center to track malicious vehicles. The formal security proof and comparative analysis validate that our protocol preserves higher security and lower overhead by comparison with related schemes.

Index Terms—Authentication protocol, privacy protection, signature, traffic accident handling, vehicle to vehicle (V2V), vehicular ad-hoc networks (VANETs).

I. INTRODUCTION

VEHICULAR ad-hoc networks (VANETs) refer to the effective utilization of vehicle environment and dynamic information through on-board unit (OBU), roadside unit (RSU), and other equipment with the help of emerging vehicle communication technology. Not only can VANETs realize network connection and information exchange between V2V, vehicle to infrastructure (V2I), vehicle to pedestrian, etc., but also provide users with safe, comfortable, intelligent, and efficient transportation services. Due to the inevitability of

traffic accidents, the wide applications of VANETs play a positive role in generating accident reports, aftercare work, and secondary disaster prevention.

Generally speaking, when a traffic accident occurs between vehicles, both parties need to authenticate each other, read the accident scene from the on-board instrument, and then jointly sign it to prevent repudiation and facilitate subsequent processing. It is a challenging task to design an efficient traffic accident handling protocol for mobile vehicles in VANETs fulfilling various security requirements.

He et al. [1] proposed a V2V and V2I authentication protocol with batch verification. Mukherjee et al. [2] designed an RSU-led report online and offline distribution protocol, the system provides subscription services for VANETs. Azees et al. [3] proposed an anonymous authentication scheme, which achieves misbehaving vehicle tracking and cross-domain authentication with different RSUs. Cui et al. [4] proposed a cuckoo filter and binary search-based authentication scheme for support of batch verification. Liu et al. [5] designed a conditional privacy-preserving vehicle announcement protocol. the initiated vehicle generates an accident report and broadcasts it to other vehicles for witness and signature. Then the initiator generates an aggregation packet and sends it to RSU. RSU judges the authenticity of the message according to the reputation value of the vehicle. Chen et al. [6] proposed a vehicle emergency event-sharing protocol, the vehicle sends emergency information via RSU to the broadcasting center, which issues an emergency certificate to the vehicle and broadcasts the emergency message through RSUs if the message is valid. Zhang et al. [7] proposed a group communication authentication protocol for VANETs using the Chinese Remainder Theorem, but Xiong et al. [8] pointed out that their scheme suffers from impersonation attack. Xie and Huang [9] proposed a secure V2I authentication protocol for VANETs, which uses the physical unclonable function (PUF) to resist RSU captured attacks, and designed a three-factor secrecy strategy to resist side channel attacks. Cheng et al. [10] integrated homomorphic encryption and data aggregation technology to propose a vehicle feedback protocol. RSU evaluates the reputation score of the feedback sender and then verifies and aggregates the feedback. The aggregated result is transmitted to the cloud service provider, which decrypts the result and calculates the reputation scores of vehicles. Mei et al. [11] proposed an authentication protocol for transportation cyber-physical system. RSU receives the traffic information sent by the vehicle and stores it in the edge server, the adjacent RSUs form the edge service domain.

Manuscript received 25 February 2024; revised 10 April 2024 and 16 May 2024; accepted 11 June 2024. Date of publication 14 June 2024; date of current version 25 September 2024. This work was supported in part by the Hangzhou Joint Fund of the Zhejiang Provincial Natural Science Foundation of China under Grant LHZSZ24F020002, and in part by the National Natural Science Foundation of China under Grant U21A20466. (Corresponding authors: Qi Xie; Xiao Tan.)

Qi Xie, Zixuan Ding, Qingyun Xie, Xiao Tan, and Wen Tang are with the Key Laboratory of Cryptography of Zhejiang Province, Hangzhou Normal University, Hangzhou 311121, China (e-mail: qixie68@126.com; dingzixuan8899@163.com; qingyunxie_m@163.com; xiaotan_cs@163.com; tangwen1012@163.com).

Debiao He is with the School of Cyber Science and Engineering, Wuhan University, Wuhan 430072, China (e-mail: hedebiao@163.com).

Digital Object Identifier 10.1109/IIOT.2024.3414453

Chen et al. [12] proposed a vehicle emergency report protocol. After witnessing an accident, vehicles report the message to the RSU, the RSU broadcasts the emergency message to seek the verification of the accident. The vehicles generate a multisignature and send it to the RSU. If the signature passes the RSU's verification, the accident report will be uploaded to the transportation authorities. Baee et al. [13] integrated message authentication and beacon encryption to present a vehicle broadcast authentication protocol, which can be used for V2I and V2V broadcast communication. Xie et al. [14] proposed a lightweight and anonymous authentication protocol for VANETs, which achieves both V2I and V2V authentication.

Blockchain originated from Satoshi Nakamoto's Bitcoin [15] in 2008. As the underlying technology of Bitcoin, blockchain is essentially a distributed ledger and new application mode of computer technology, such as distributed data storage, point-to-point transmission, consensus mechanism, and encryption algorithm. Blockchain technology is a technical solution that does not rely on a third party and stores, verifies, transmits, and communicates through its decentralized nodes. As a decentralized, tamper-proof, and nonrepudiation emerging technology, blockchain is also widely used in the VANETs. Li et al. [16] proposed homomorphic encryption and blockchain-based V2R and V2V authentication protocol. However, it requires more computational cost. Zhang and Xu [17] presented an anonymous authentication protocol based on the certificateless message signature and blockchain incentives, but adversaries can seek rewards or escape punishment by collusion attacks. Xie et al. [18] proposed a blockchain-based V2I handover authentication, and V2V broadcasting authentication protocol, which can realize V2V broadcast traffic information. To realize decentralized data storage, Qureshi et al. [19] proposed a blockchain-based authentication model, which uses smart contracts and RSU to authenticate the vehicles. To address the issue of malicious vehicles spreading erroneous information, Zhao et al. [20] proposed a trust management model based on blockchain, machine learning, and active detection technologies. Roy et al. [21] proposed a blockchain-based access control protocol with handover authentication. Du et al. [22] proposed a trust value management method based on consortium blockchain to establish a trustworthy vehicle information sharing environment. Wei et al. [23] proposed a conditional privacy-preserving authenticated key agreement scheme with multi-TA model for fog-based VANETs. Xie et al. [24] proposed a blockchain-based cross-trusted authority authentication protocol for VANETs. Alharthi et al. [25] proposed a blockchain-based method for calculating vehicle credibility and subsequently predicting vehicle credibility.

A. Motivations and Contributions

The previous traffic accident handling schemes relied on witness vehicles to send the accident report to RSU, which broadcasts the report to find other witness vehicles. RSU judges authenticity of the report by aggregated signature or

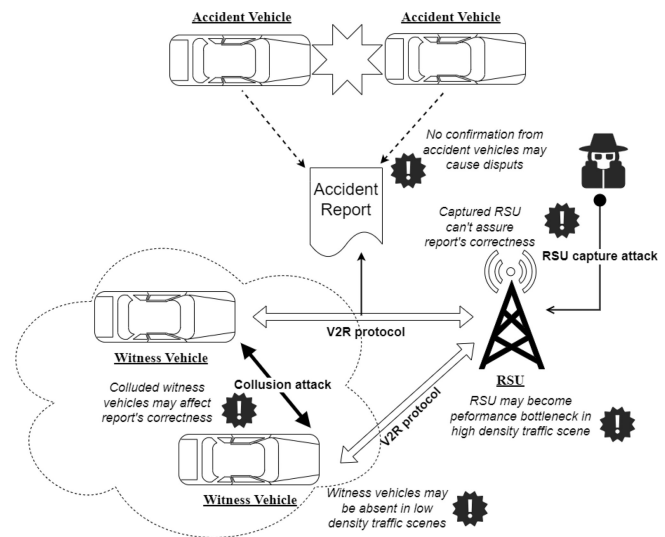


Fig. 1. Security vulnerabilities in existing schemes for VANETs.

reputation value of the witness vehicles. Under this situation, there are some weaknesses: 1) because the report is not confirmed by the involved vehicles, it will probably cause disputes after the accident; 2) in cases where there lack RSU deployment or RSU is captured, existing schemes will no longer work or fail to preserve correctness of the report; 3) since the aggregated signature is generated by the witness vehicles, it may suffer from collusion attacks; and 4) it is not scalable to the low-density traffic scenario where witness vehicles are unavailable or the high-density scenario where RSU may become the bottleneck of performance. Fig. 1 shows those security vulnerabilities in existing schemes for VANETs.

Therefore, we propose a novel V2V and V2I authentication and traffic accident handling protocol, namely, accident processing authentication (APA) protocol, to address the above issues. The contributions of this article are listed as follows.

- 1) We propose a novel self-sufficient traffic accident handling mechanism, such that the generation of traffic accident reports relies solely on the vehicles involved in accident, eliminating the dependence on RSUs or witness vehicles. This approach enhances autonomy of the vehicular communication system and reduces reliance on external infrastructure.
- 2) Blockchain technology is adopted in our system model to keep jointly signed accident reports, so as to enforce integrity and traceability of reports. Besides, registration information of vehicles and RSUs are also inscribed on the blockchain and retrievable as credentials from the registration center (RC), for realizing efficient authentications in V2V and V2R protocols.
- 3) A dynamic pseudo-identity strategy is devised by integrating symmetric encryption with elliptic curve cryptosystem (ECC), for ensuring unlinkability of pseudo-identities and preserve privacy of vehicles, while still allowing the RC to track malicious vehicles effectively.
- 4) Formal proofs are conducted for validating the correctness and security of our protocol. Through analysis and

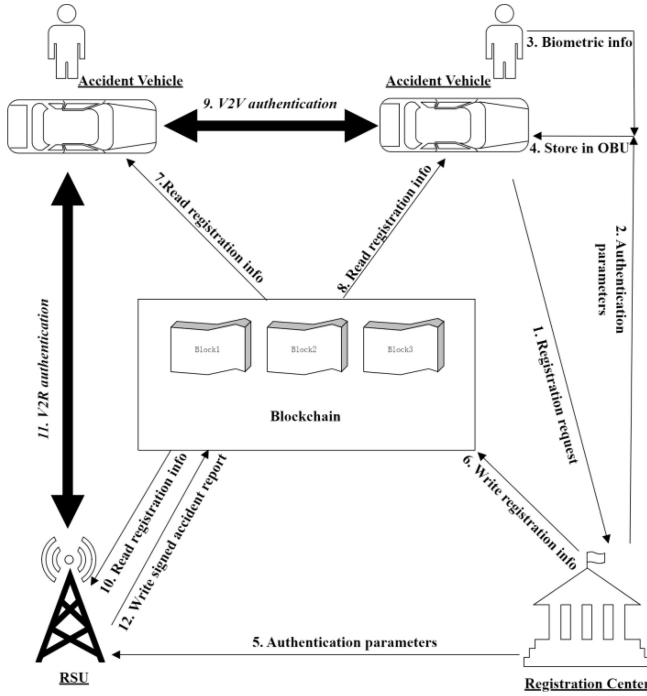


Fig. 2. System model.

comparison with related schemes, our scheme outperforms them regarding both efficiency and many security properties.

In Section II, we introduce the system model and design goals. The adversary model and the assumptions are presented in Section III. We propose a novel authentication and signature protocol in Section IV. Section V shows the formal proof and security analysis of the protocol. Section VI conducts a simulation experiment on the proposed protocol by comparison to related schemes. This article concludes in Section VII.

II. SYSTEM MODEL AND DESIGN GOALS

A. System Model

The system model is shown in Fig. 2, which covers the registration phase and accident report generation phase.

In the registration phase, the driver and vehicle send registration request (message flow 1) to the RC. After the identity verification, RC returns the authentication parameters (message flow 2), and the driver inputs the bio-information (message flow 3) to encrypt them and store in the vehicle's OBU (message flow 4). RSU also receives its authentication parameters from RC and stores them (message flow 5). Besides, RC is authorized to write partial registration information of vehicles and RSUs onto the blockchain (message flow 6).

In the accident report generation phase, the accident vehicles first retrieve their registration information from the blockchain (message flows 7 and 8) and then authenticate each other using V2V protocol (message flow 9) to generate the signed accident report. Upon entry into the RSU domain, either of the accident vehicles can notify RSU, which first reads its registration information from the blockchain (message

flow 10) and then runs V2R protocol with the vehicle for authentication (message flow 11). Finally, RSU writes the jointly signed accident report to the blockchain (message flow 12).

Notice that access control policy is enforced upon the blockchain. In particular, the authorized RC and RSU have written permission to the blockchain regarding registration information and signed accident reports, respectively, while the other relevant entities (e.g., vehicles, traffic police, and insurance companies) are only granted read permission to the blockchain. Outsiders of the system have neither read nor write permission to the blockchain for protection of personal data privacy.

B. Design Goals

The design goals of our scheme are listed as follows: 1) even if a traffic accident occurs in a place without traffic infrastructure, the traffic accident report can only be generated by relying on each accident vehicle, and the traffic accident report has the signature of each accident vehicle; 2) the signature of each accident vehicle in the traffic accident report can be verified by RSU and written onto the blockchain by RSU, which is convenient for the traffic police or insurance company to handle the accident, and has nonrepudiation and public verifiability; 3) the proposed protocol can protect the privacy of vehicle, support RC's tracking of malicious messages, and resist all known attacks, including OBU intrusion attack, RSU capture attack, etc.; and 4) the proposed traffic accident handling scheme can be proved to be secure and efficient.

III. ASSUMPTIONS AND ADVERSARIAL MODEL

A. Assumptions

The provable security of our scheme is based on the intractability of the elliptic curve discrete logarithm problem (ECDLP) assumption and semantic security of symmetric encryption scheme (chosen ciphertext attack, CCA), defined as follows.

Definition 1 (ECDLP Assumption): Given a generator $P \in E(F_p)$ of order n and one point $A = aP$, no probabilistic polynomial time (PPT) adversary can find a with nonnegligible probability.

Definition 2 (Semantic Security of Symmetric Encryption): A symmetric encryption scheme $SE(\cdot) = \{E_k(\cdot), D_k(\cdot)\}$ provides semantic security (or CCA security), if any PPT adversary $\mathcal{A}$ playing the following game with a challenger $\mathcal{C}$.

Phase 1: $\mathcal{A}$ adaptively queries to encryption oracle $E_k(\cdot)$ and decryption oracle $D_k(\cdot)$.

Phase 2: $\mathcal{A}$ chooses two messages m_0 and m_1 , submits to $\mathcal{C}$. $\mathcal{C}$ tosses a bit $b \leftarrow \{0, 1\}$ and returns $c^* = E_k(m_b)$ to $\mathcal{A}$.

Phase 3: $\mathcal{A}$ adaptively queries to $E_k(\cdot)$ and $D_k(\cdot)$, under the restriction that c^* is not queried to $D_k(\cdot)$. Finally, $\mathcal{A}$ outputs a bit b' .

The probability that $b' = b$ is no greater than $1/2$, says the advantage of $\mathcal{A}$

$$\text{Adv}_{\text{CCA}}(\mathcal{A}) = 2 \Pr[b' = b] - 1$$

is negligible.

B. Threat Model

The capabilities of an adversary we consider are listed as follows.

C1: An adversary can eavesdrop on any message over public channels.

C2: An adversary can insert, delete, modify, or replay any message over public channels.

C3: An adversary can corrupt the private keys of vehicles or RSUs.

C4: An adversary can steal the data in OBU.

C5: An adversary can fake the biometrics of drivers.

Especially, C1 models passive adversaries, C2 models active adversaries, C3 models insider adversaries, and C4 and C5 models two-factor authentication.

C. Adversarial Model

The adversarial model of our anonymous two-factor APA protocol is defined as follows. There are three types of participants: 1) a trusted RC RC ; 2) n vehicles $V = \{V_i | i = 1, n\}$; and 3) m RSU $R = \{R_t | t = 1, m\}$. Each vehicle V_i and each RSU R_t have unique identities VID_i and RID_t , respectively. Let Π_α^x denote the x th instance of the participant $\alpha \in V \cup R$. We say that Π_α^x and Π_β^y are partners if they can authenticate mutually and generate signatures on the co-related accident record.

Definition 3 (Correctness): The transaction data $\{c_{ij}, PK_{V_i}, PK_{V_j}, AC1, AC2, D_{ij}, T_1, PID_i, T_2, PID_j, Val_i, Val_j, RID_t, PK_{R_t}, T_4, c_t, D_t\}$ uploaded to the blockchain is said to be correct if the following conditions hold.

- 1) $c_{ij} \cdot P = PK_{V_j} + PK_{V_i} + h(AC1 \| AC2 \| D_{ij} \| T_1 \| PID_i \| T_2 \| PID_j \| Val_i \| Val_j) \cdot D_{ij}$.
- 2) $c_t \cdot P = PK_{R_t} + h(c_{ij} \| T_4 \| D_t) \cdot D_t$.

Definition 4 (Existential Unforgeability): We assume that an adversary $\mathcal{A}$ against existential unforgeability is a PPT algorithm that can control all public communications and access to a set of oracles: hash, execute, send, extract, and corrupt, defined as below.

Hash(m): This oracle returns the hash output $h(m)$ when a message m is queried.

Execute($\Pi_\alpha^x, \Pi_\beta^y$): This oracle returns the protocol transcript running between Π_α^x and Π_β^y . It is to model the adversarial capability C1.

Send($\Pi_\alpha^x, \cdot$): This oracle returns the response by Π_α^x . $\cdot$ denotes the auxiliary input to specify the protocol phase and the previous message to respond. It is to model the adversarial capability C2.

Extract($ID, \cdot$): This oracle returns the private key associated with ID . $\cdot$ denotes the auxiliary input depending on which type of participant (vehicle or RSU) that ID belongs to. It is to model the adversarial capability C3.

Corrupt($V_i, \cdot$): This oracle returns the OBU data or biometric of V_i . $\cdot$ denotes the auxiliary input to specify the factor of authentication. It is to model the adversarial capabilities C4 and C5.

We say that an APA protocol achieves Existential Unforgeability if $\mathcal{A}$ has negligible probability of producing

correct transaction data: $\{c_{ij}^*, PK_{V_i}^*, PK_{V_j}^*, AC1, AC2, D_{ij}^*, T_1, PID_i^*, T_2, PID_j^*, Val_i^*, Val_j^*, RID_t^*, PK_{R_t}^*, T_4, c_t^*, D_t^*\}$.

Under the following restrictions:

- 1) $\mathcal{A}$ never made execute and send queries related to $\Pi_{V_i}^x, \Pi_{V_j}^y, \Pi_{R_t}^z$;
- 2) $\mathcal{A}$ never made extract queries related to $PID_i^*, PID_j^*, RID_t^*$;
- 3) $\mathcal{A}$ never made corrupt queries related to V_i^*, V_j^* in both OBU and biometric.

Definition 5 (Anonymity): We assume that an adversary $\mathcal{A}$ against anonymity is a PPT algorithm that can control all public communications and access to a set of oracles: hash, execute, send, extract, and corrupt, defined similarly as in Definition 4; reveal and test, defined as below.

Reveal($\Pi_{V_i}^x$): This oracle returns the real identity VID_i of V_i .

Test($\Pi_{V_i}^x$): This oracle can be queried only once. When queried, it tosses a bit $b \leftarrow \{0, 1\}$ and returns the real identity VID_i of V_i if $b = 1$, or returns a random value in the identity space if $b = 0$.

We say that an APA protocol achieves anonymity if the probability that $\mathcal{A}$ correctly guesses b is no greater than $1/2$ under the restriction that $\mathcal{A}$ never queried $\Pi_{V_i}^x$ to Reveal oracle where $\Pi_{V_i}^x$ is the query $\mathcal{A}$ made to Test oracle.

IV. PROPOSED SCHEME

A novel vehicle traffic accident handling protocol, namely, APA protocol, is proposed in this section. Tables I and II, respectively, summarize the notations and acronyms used in this article.

Fig. 3 shows the technology roadmap of our protocol. By integrating symmetric encryption and ECC, we develop a dynamic pseudo-identity strategy to enforce vehicle identity privacy. Blockchain technology is utilized to achieve accident integrity and traceability. Two-factor authentication, aggregatable signature, and lightweight cryptography are integrated to realize secure and efficient V2V and V2R authentication. Based on the above components, our APA protocol is designed delicately with solid foundation and high reliability.

A. Initialization

RC chooses an elliptic curve $E(F_p)$ with a generator P , a symmetric encryption scheme $SE(\cdot) = \{E_k(\cdot), D_k(\cdot)\}$, a one-way hash function $h(\cdot)$, the reproduction and generation functions $Gen(\cdot)$, $Rep(\cdot)$ of a fuzzy extractor algorithm, a long-term secret random number K_{TA} . RC publishes $\{P, E(F_p), SE(\cdot), h(\cdot), Gen(\cdot), Rep(\cdot)\}$, and we assume that those global parameters are accessible to all the entities in the system.

B. Registration Phase

The vehicles and RSUs register with RC in the registration phase, illustrated as follows.

Vehicle Registration:

Step VR1: The driver inputs the biometrical information Bio_i and the vehicle V_i selects a secret random number SK_{V_i} .

TABLE I
NOTATIONS

Notations	Description
V_i	i -th vehicle
TA	The trusted-authority
RSU_t	t -th roadside unit
Bio_i	The biological information of V_i 's driver
VID_i	Unique identification (engine number) of V_i
Val_i	The appearance characteristics of V_i
PID_i	V_i 's temporary pseudo-identity
K_{TA}	TA's secret key
P	Generator point of elliptic curve
$SE(\cdot)$	Symmetric encryption scheme
$h(\cdot)$	Hash function
$\parallel$	Concatenation
$\oplus$	XOR operation
$Rep(\cdot), Gen(\cdot)$	Fuzzy extractor algorithm for reproduction and generation
τ_i	Reproduction parameter of fuzzy extractor algorithm
σ_i	Driver's biometric key
$PUF()$	Physical unclonable function
Cha_t, Res_t	The challenge and response of the $PUF()$
T_1, T_2, T_3	Timestamps
ΔT	The maximum transmission delay time
$AC1, AC2$	Accident record

TABLE II
ACRONYMS

Acronyms	Description
VANET	Vehicular Ad-hoc Network
V2V	Vehicle to Vehicle
V2R	Vehicle to Roadside Unit
V2I	Vehicle to Infrastructure
OBU	On-Board Unit
RSU	Roadside Unit
RC	Registration Center
APA	Accident Processing Authentication
ECC	Elliptic Curve Cryptosystem
ECDLP	Elliptic Curve Discrete Logarithm Problem
CCA	Chosen Ciphertext Attack

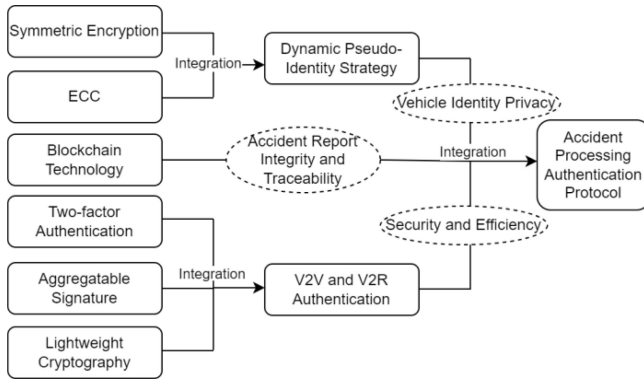


Fig. 3. Technology roadmap of proposed protocol.

V_i computes $PK_{vi} = SK_{vi} \cdot P$ and sends $\{VID_i, PK_{vi}, Val_i\}$ to RC via the secure channel, where P is the generator point, VID_i and Val_i are the vehicle's engine number and appearance characteristics, respectively.

Step VR2: On receiving $\{VID_i, PK_{vi}, Val_i\}$, RC verifies the uniqueness and the legitimacy of VID_i and PK_{vi} , if

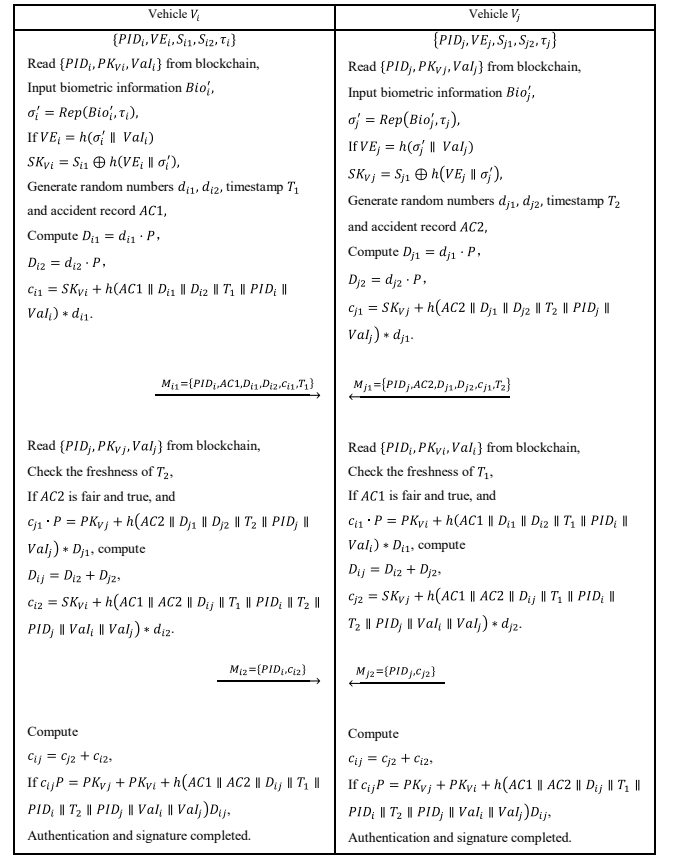


Fig. 4. V2V authentication and accident report generation.

not, RC rejects the request, else generates random number r_i , then computes $PID_i = E_{K_{TA}}(VID_i, Val_i, r_i)$, and writes $\{PID_i, PK_{vi}, Val_i\}$ onto the blockchain, and sends $\{PID_i, r_i\}$ to the vehicle via the secure channel.

Step VR3: On receiving $\{r_i\}$, the driver inputs his/her biological information Bio_i , the vehicle computes $(\sigma_i, \tau_i) = Gen(Bio_i)$, $VE_i = h(\sigma_i \parallel Val_i)$, $S_{i1} = SK_{vi} \oplus h(VE_i \parallel \sigma_i)$, $S_{i2} = r_i \oplus h(VID_i \parallel \sigma_i)$. Then the parameters $\{PID_i, VID_i, VE_i, S_{i1}, S_{i2}, \tau_i\}$ are stored in the vehicle's OBU.

RSU Registration: RC selects a unique RSU identity RID_t , a secret random number SK_{Rt} , then computes $PK_{Rt} = SK_{Rt} \cdot P$, and uploads $\{RID_t, PK_{Rt}\}$ onto the blockchain, and sends $\{RID_t, SK_{Rt}\}$ to RSU_t via a secure channel. RSU_t generates a challenge Cha_t , computes $Res_t = PUF(Cha_t)$, $RSK_{Rt} = Res_t \oplus SK_{Rt}$, and stores $\{RID_t, RSK_{Rt}, PUF(), Cha_t\}$.

C. V2V Authentication and Traffic Accident Report Generation

This part describes the process of mutual authentication between vehicles after an accident and the joint signature of accident reports. The steps are described in Fig. 4.

Step VAI: Suppose a traffic accident occurs between the vehicles V_i and V_j , the drivers input their biometrical information. Take V_i 's driver as an example: V_i first read the registration information $\{PID_i, PK_{vi}, Val_i\}$ from blockchain according to PID_i ; after inputting the biometrical information Bio'_i , V_i computes $\sigma'_i = Rep(Bio'_i, \tau_i)$

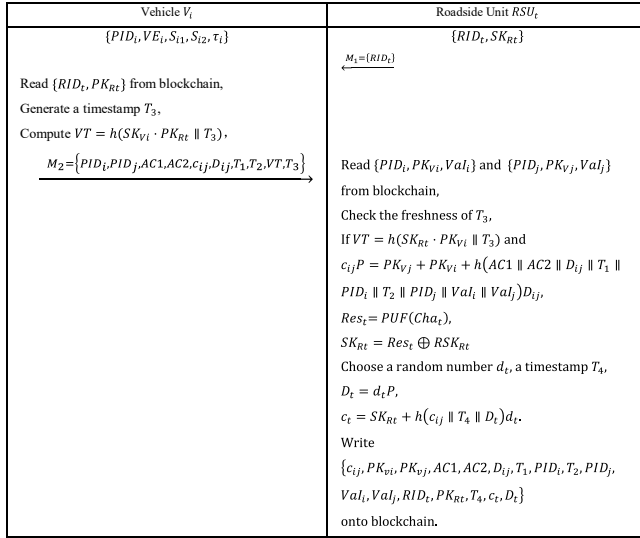


Fig. 5. V2R authentication and accident report upload.

and checks if $VE_i = h(\sigma'_i \parallel Val_i)$, if not, aborts it, else computes $SK_{Vi} = S_{i1} \oplus h(VE_i \parallel \sigma'_i)$ to recover the secret random number SK_{Vi} . Then V_i generates random numbers d_{i1} , d_{i2} , timestamp T_1 , and the accident record $AC1$ read from the on-board instrument, computes $D_{i1} = d_{i1} \cdot P$, $D_{i2} = d_{i2} \cdot P$, $c_{i1} = SK_{Vi} + h(AC1 \parallel D_{i1} \parallel D_{i2} \parallel T_1 \parallel PID_i \parallel Val_i) \cdot d_{i1}$, and sends $M_{i1} : \{PID_i, AC1, D_{i1}, D_{i2}, c_{i1}, T_1\}$ to the vehicle V_j . Similarly, the vehicle V_j generates and sends $M_{j1} : \{PID_j, AC2, D_{j1}, D_{j2}, c_{j1}, T_2\}$ to the vehicle V_i via the public channel.

Step VA2: On receiving M_{j1} , V_i first read the registration information $\{PID_j, PK_{Vj}, Val_j\}$ from blockchain according to PID_j , checks if $|T_2^* - T_2| < \Delta T$, and accident record $AC2$ is correct and fair, if not, V_i aborts the process, else V_i checks the integrity of the message by verifying if $c_{j1} \cdot P = PK_{Vj} + h(AC2 \parallel D_{j1} \parallel D_{j2} \parallel T_2 \parallel PID_j \parallel Val_j) \cdot D_{j1}$, if the message M_{j1} passes the verification, V_i computes $D_{ij} = D_{i2} + D_{j2}$, $c_{i2} = SK_{Vi} + h(AC1 \parallel AC2 \parallel D_{ij} \parallel T_1 \parallel PID_i \parallel T_2 \parallel PID_j \parallel Val_i \parallel Val_j) \cdot d_{i2}$, and sends $M_{i2} : \{PID_i, c_{i2}\}$ to V_j via the public channel. Similarly, after verifying the legitimacy of V_i 's identity and the integrity of the message M_{i1} , V_j generates and sends the message $M_{j2} : \{PID_j, c_{j2}\}$ to V_i publicly.

Step VA3: On receiving the message sent by the other party, the vehicles integrate the accident report by computing $c_{ij} = c_{j2} + c_{i2}$, then verify if $c_{ij} \cdot P = PK_{Vj} + PK_{Vi} + h(AC1 \parallel AC2 \parallel D_{ij} \parallel T_1 \parallel PID_i \parallel T_2 \parallel PID_j \parallel Val_i \parallel Val_j) \cdot D_{ij}$, if not, abort the process, else authentication and signature completed.

Note that the proposed solution is applicable to the handling of traffic accident reports involving multiple vehicles simultaneously involved in traffic accidents.

D. V2R Authentication and Traffic Accident Report Upload

After mutual authentication and joint signature of the accident report, the vehicles leave the scene and upload the accident report to the blockchain via RSU by running the V2R authentication protocol shown in Fig. 5, where we assume

that vehicle V_i still keeps the results in the execution of V2V authentication protocol.

Step RA1: The vehicle V_i receives the broadcast message $M1 : \{RID_t\}$ from RSU_t , read the registration information $\{RID_t, PK_{Rt}\}$ from blockchain according to RID_t , generates a timestamp T_3 and computes $VT = h(SK_{Vi} \cdot PK_{Rt} \parallel T_3)$. Then, V_i sends the message $M2 : \{PID_i, PID_j, AC1, AC2, c_{ij}, D_{ij}, T_1, T_2, VT, T_3\}$ to RSU_t via the public channel.

Step RA2: On receiving the message $M2$, RSU_t read the registration information $\{PID_i, PK_{Vi}, Val_i\}$ and $\{PID_j, PK_{Vj}, Val_j\}$ from blockchain according to PID_i and PID_j , checks if T_3 is fresh and verifies if $h(SK_{Rt} \cdot PK_{Vi} \parallel T_3) = VT$ and $c_{ij} \cdot P = PK_{Vj} + PK_{Vi} + h(AC1 \parallel AC2 \parallel D_{ij} \parallel T_1 \parallel PID_i \parallel T_2 \parallel PID_j \parallel Val_i \parallel Val_j) \cdot D_{ij}$, if not, RSU_t aborts it, else generates a random number d_t , a timestamp T_4 , and computes $Res_t = PUF(Cha_t)$, $SK_{Rt} = Res_t \oplus RSK_{Rt}$, $D_t = d_t \cdot P$, $c_t = SK_{Rt} + h(c_{ij} \parallel T_4 \parallel D_t) \cdot d_t$. Then, RSU_t writes $\{c_{ij}, PK_{Vi}, PK_{Vj}, AC1, AC2, D_{ij}, T_1, PID_i, T_2, PID_j, Val_i, Val_j, RID_t, PK_{Rt}, T_4, c_t, D_t\}$ onto the blockchain.

In case of dispute, RSU sends PID_i and PID_j to RC, RC can decrypt them and obtain the engine numbers and appearance characteristics of vehicles.

E. Vehicle Temporary Identity Update

The vehicle can update the temporary identity and identity parameters to resist tracking attacks and privacy disclosure.

Step VT1: V_i 's driver inputs the biometrical information Bio'_i , V_i computes $\sigma'_i = Rep(Bio'_i, \tau_i)$ and verifies if $VE_i = h(\sigma'_i \parallel Val_i)$, if not, aborts the process, else, V_i computes $r_i = S_{i2} \oplus h(VID_i \parallel \sigma'_i)$ to recover the secret parameter r_i . Then, V_i selects a secret random number SK_{Vi}^* , a timestamp T_5 , and computes $PK_{Vi}^* = SK_{Vi}^* \cdot P$, $M_1 = E_{r_i}(VID_i, PK_{Vi}^*, PID_i, T_5)$. Then, V_i sends $\{PID_i, M_1, T_5\}$ to RC via the public channel.

Step VT2: On receiving the message $\{PID_i, M_1, T_5\}$, RC first checks the freshness of T_5 , and computes $(VID_i, Val_i, r_i) = D_{K_{TA}}(PID_i)$, $(VID'_i, PK_{Vi}^*, PID'_i, T'_5) = D_{r_i}(M_1)$. RC verifies if $VID'_i = VID_i$, $PID'_i = PID_i$, and $T'_5 = T_5$, if not, RC terminates the session, else, generates random numbers r_i^* , a timestamp T_6 , and computes $PID_i^* = E_{K_{TA}}(VID_i \parallel Val_i \parallel r_i^*)$, $M_2 = E_{r_i}(VID_i, PID_i^*, r_i^*, T_6)$. Then, RC deletes $\{PID_i, PK_{Vi}, Val_i\}$ from the blockchain, writes $\{PID_i^*, PK_{Vi}^*, Val_i\}$ onto the blockchain, and sends $\{PID_i, M_2, T_6\}$ to V_i via the public channel.

Step VT3: On receiving $\{PID_i, M_2, T_6\}$, V_i first checks the freshness of T_6 , and computes $(VID'_i, PID_i^*, r_i^*, T'_6) = D_{r_i}(M_2)$. If $VID'_i = VID_i$, and $T'_6 = T_6$, V_i computes $S_{i1}^* = SK_{Vi}^* \oplus h(VE_i \parallel \sigma_i)$, $S_{i2}^* = r_i^* \oplus h(VID_i \parallel \sigma_i)$, and updates $\{PID_i^*, VE_i, S_{i1}^*, S_{i2}^*, \tau_i\}$ in the OBU.

V. SECURITY ANALYSIS

In this section, we will formally prove the security of the proposed scheme under the random oracle model, and then provide informal cryptanalysis to show that the proposed scheme achieves several desirable security properties.

A. Security Proofs

The correctness of our proposed APA protocol is straightforward, therefore is omitted here.

Theorem 1: The proposed APA protocol achieves existential unforgeability under ECDLP assumption.

Proof: Suppose there exists a PPT adversary $\mathcal{A}$ which can break the existential unforgeability of the proposed protocol with nonnegligible probability ϵ , by producing a transaction data Tx^* in which the involved participants are V_i^* , V_j^* , R_t^* . We construct a simulator $\mathcal{B}$ to simulate the oracles, such that given an ECDLP instance $(P, E(F_p), A = aP)$, $\mathcal{B}$ can find a with nonnegligible probability.

Without loss of generality, assume that $\mathcal{A}$ never made queries corresponding to V_i^* , including $Extract(VID_i^*, pk_{Vi}^*, Val_i^*)$, $Execute(\Pi_{V_i^*}^x, \Pi_{V_j}^y)$, $Execute(\Pi_{V_i^*}^x, \Pi_{R_t}^z)$, $Send(\Pi_{V_i^*}^x, 1, start)$, $Send(\Pi_{V_i^*}^x, 1, < D2, b_j, PID_j, A_j, D_{j1}, D_{j2}, PK_{Vj}, c_{j1}, Val_j, T_2 >)$, $Send(\Pi_{V_i^*}^x, 2, < c_{j2}, PID_j, Val_j, PID_i >)$, $Send(\Pi_{V_i^*}^x, 2, < RID_t, y_t, Z_t, PK_{Rt} >)$, $Corrupt(V_i^*, OBU) \& Corrupt(V_i^*, BIO)$ where we use 1 to denote the accident record confirming and signing phase, and 2 to denote the vehicle & RSU authentication and signature verification phase in send queries.

Then $\mathcal{B}$ follows the computations in the system initialization phase, vehicles registration phase, and RSUs registration phase, to generate and store the following data.

- 1) Public parameters $Para = \{P, E(F_p), SE(\cdot), h(\cdot), Gen(\cdot), Rep(\cdot)\}$ where $P, E(F_p)$ are from the ECDLP instance.
- 2) $\{PID_i, SK_{Vi}, PK_{Vi}, Val_i, VID_i, Bio_i, (VE_i, S_{i1}, S_{i2}, \tau_i)\}_{i=1, \dots, n}$ for individual data of the n vehicles where $i \neq i^*$, and $\{PID_{i^*}, PK_{Vi^*}, Val_{i^*}, VID_{i^*}, Bio_{i^*}, (VE_{i^*}, S_{i^*1}, S_{i^*2}, \tau_{i^*})\}$, where $i = i^*$. Here, we denote this list as $\mathcal{L}_{Vi}$, and set $PK_{Vi^*} = A$.
- 3) $\{RID_t, SK_{Rt}, PK_{Rt}\}_{t=1, \dots, m}$ for registration data of the m RSUs. Here, we denote this list as $\mathcal{L}_{Rt}$.

The oracles are simulated by $\mathcal{B}$ in the following way.

Hash(m): $\mathcal{B}$ maintains a list $\mathcal{L}_h = \{(m, r)\}$ to answer the hash queries. Given an input m , if (m, r) is already in $\mathcal{L}_h$, return r ; otherwise, pick a random r from the range of hash function $h(\cdot)$ with no collision to previous queries, insert (m, r) into $\mathcal{L}_h$ and return r .

Execute($\Pi_{\alpha}^x, \Pi_{\beta}^y$): There are two types of execute queries, according to whether the protocol is running between V_i and V_j or between V_i and R_t .

Execute($\Pi_{V_i}^x, \Pi_{V_j}^y$): $\mathcal{B}$ returns the protocol transcript $\{< PID_i, AC1, D_{i1}, D_{i2}, c_{i1}, T_1 >, < PID_j, AC2, D_{j1}, D_{j2}, c_{j1}, T_2 >, < PID_i, c_{i2} >, < PID_j, c_{j2} >\}$ using four send queries as:

$Send(\Pi_{V_i}^x, 1, start)$, $Send(\Pi_{V_j}^y, 1, start)$
 $Send(\Pi_{V_i}^x, 1, < PID_j, AC2, D_{j1}, D_{j2}, c_{j1}, T_2 >)$
 $Send(\Pi_{V_j}^y, 1, < PID_i, AC1, D_{i1}, D_{i2}, c_{i1}, T_1 >)$.

Execute($\Pi_{V_i}^x, \Pi_{R_t}^z$): $\mathcal{B}$ returns the protocol transcript $\{< RID_t >, < PID_i, PID_j, AC1, AC2, c_{ij}, D_{ij}, T_1, T_2, VT, T_3 >, \text{using two send queries as: } Send(\Pi_{R_t}^z, 2, start), Send(\Pi_{V_i}^x, 2, < RID_t >)\}$.

Send($\Pi_{\alpha}^x, \cdot$): There are five types of send query, according to the message flows sent during 1) accident record confirming

and signing phase and 2) vehicle and RSU authentication and signature verification phase. Note that it is unnecessary to simulate messages sent in the other phases that are supposed to communicate through secure channels.

Send($\Pi_{V_i}^x, 1, start$): $\mathcal{B}$ computes $D_{i1} = d_{i1} \cdot P$, $D_{i2} = d_{i2} \cdot P$ where d_{i1} and d_{i2} are random values from Z_p , and $c_{i1} = SK_{Vi} + h(AC1 \| D_{i1} \| D_{i2} \| T_1 \| PID_i \| Val_i) * d_{i1}$ where $AC1$ is the accident record and T_1 is the current time stamp, then stores $< AC1, T_1, d_{i2}, D_{i2} >$ as $\Pi_{V_i}^x$'s state (every instance may have a temporary state during its lifecycle), returns $< PID_i, AC1, D_{i1}, D_{i2}, c_{i1}, T_1 >$ as the first-round message of phase (1). Note that $PID_i, PK_{Vi}, SK_{Vi}, Val_i$ are read from $\mathcal{L}_{Vi}$.

Send($\Pi_{V_i}^x, 1, < PID_j, AC2, D_{j1}, D_{j2}, c_{j1}, T_2 >$): $\mathcal{B}$ checks the freshness of T_2 , verifies whether $c_{j1} \cdot P = PK_{Vj} + h(AC2 \| D_{j1} \| D_{j2} \| T_2 \| PID_j \| Val_j) * D_{j1}$ holds, returns $\perp$ if verification failed. Otherwise, $\mathcal{B}$ uses $\Pi_{V_i}^x$'s state to compute $D_{ij} = D_{i2} + D_{j2}$, $c_{i2} = SK_{Vi} + h(AC1 \| AC2 \| T_1 \| PID_i \| T_2 \| PID_j \| Val_i \| Val_j \| D_{ij}) * d_{i2}$, refreshes $\Pi_{V_i}^x$'s state as $< AC1, AC2, T_1, T_2, D_{ij}, c_{i2} >$, and returns $< PID_i, c_{i2} >$ as the second-round message of phase (1).

Send($\Pi_{R_t}^z, 2, start$): $\mathcal{B}$ returns $< RID_t >$ read from $\mathcal{L}_{Rt}$ as the first-round message of phase (2).

Send($\Pi_{V_i}^x, 2, < RID_t >$): $\mathcal{B}$ uses $\Pi_{V_i}^x$'s state to compute $c_{ij} = c_{j2} + c_{i2}$ and verify whether $c_{ij}P = PK_{Vj} + PK_{Vi} + h(AC1 \| AC2 \| T_1 \| PID_i \| T_2 \| PID_j \| Val_i \| Val_j \| D_{ij})D_{ij}$, and returns $\perp$ if verification failed. Otherwise, $\mathcal{B}$ refreshes $\Pi_{V_i}^x$'s state as $< AC1, AC2, T_1, T_2, D_{ij}, c_{ij} >$, generates a timestamp T_3 and computes $VT = h(SK_{Vi} \cdot PK_{Rt} \| T_3)$ and returns $< PID_i, PID_j, AC1, AC2, c_{ij}, D_{ij}, T_1, T_2, VT, T_3 >$ as the second-round message of phase (2).

Extract($ID, \cdot$): There are two types of extract query, according to whether the queried ID belongs to a vehicle or an RSU.

Extract(VID_i, PK_{Vi}, Val_i): $\mathcal{B}$ checks $\mathcal{L}_{Vi}$ to find the record containing VID_i, PK_{Vi}, Val_i , and returns the corresponding SK_{Vi} as the private key.

Extract(RID_t): $\mathcal{B}$ checks $\mathcal{L}_{Rt}$ to find the record containing RID_t , and returns the corresponding SK_{Rt} as the private key.

Corrupt($V_i, \cdot$): There are two types of corrupt query, according to whether the authentication factor is OBU or biometric.

Corrupt(V_i, OBU): $\mathcal{B}$ checks $\mathcal{L}_{Vi}$ to find the record containing V_i and returns the corresponding $(PID_i, VE_i, S_{i1}, S_{i2}, \tau_i)$ as the OBU data.

Corrupt(V_i, BIO): $\mathcal{B}$ checks $\mathcal{L}_{Vi}$ to find the record containing V_i and returns the corresponding Bio_i as biometric.

After adaptively querying to the above oracles, if $\mathcal{A}$ successfully outputs a correct transaction data $Tx^* = \{c_{i^*j^*}, PK_{Vj^*}, PK_{Vi^*}, AC1, AC2, D_{i^*j^*}, T_1, PID_{i^*}, T_2, PID_{j^*}, Val_{i^*}, Val_{j^*}, RID_{t^*}, PK_{Rt^*}, c_{t^*}, D_{t^*}\}$. Then we have: $c_{i^*j^*}P = PK_{Vj^*} + PK_{Vi^*} + h(AC1 \| AC2 \| T_1 \| PID_{i^*} \| T_2 \| PID_{j^*} \| Val_{i^*} \| Val_{j^*} \| D_{i^*j^*})D_{i^*j^*}$.

Denote $M^* = AC1 \| AC2 \| T_1 \| PID_{i^*} \| T_2 \| PID_{j^*} \| Val_{i^*} \| Val_{j^*}$, $H^* = h(AC1 \| AC2 \| T_1 \| PID_{i^*} \| T_2 \| PID_{j^*} \| Val_{i^*} \| Val_{j^*} \| D_{i^*j^*}) = h(M^* \| D_{i^*j^*})$, then $< D_{i^*j^*}, H^*, c_{i^*j^*} >$ can be treated as a digital signature on the message M^* co-signed by V_{i^*} and V_{j^*} , such that: $c_{i^*j^*}P = PK_{Vj^*} + PK_{Vi^*} + H^*D_{i^*j^*}$. It complies with

the format defined in Pointcheval and Stern's forking lemma. Therefore, by applying forking lemma, if $\mathcal{A}$ can forge such a signature in time t with probability $\epsilon > 7Q_h/2^k$ where Q_h is the number queries to the hash oracle and k is the security parameter, then within $\mathcal{B}$ can forge another valid signature $\langle D_{i^*j^*}, H^*, c'_{i^*j^*} \rangle$ in expected time $t' \leq 84480tQ_h/\epsilon$, such that:

$$c'_{i^*j^*}P = PK_{Vj^*} + PK_{Vi^*} + H^*D_{i^*j^*}.$$

By the above two equations, we have:

$$c_{i^*j^*} = SK_{Vj^*} + SK_{Vi^*} + H^*d_{i^*j^*}.$$

$$c'_{i^*j^*} = SK_{Vj^*} + SK_{Vi^*} + H^*d'_{i^*j^*},$$

where we denote $D_{i^*j^*} = d_{i^*j^*}P$ and SK_{Vi^*} can be obtained from $\mathcal{L}_{Vi}$ for $i = i^*$. Hence:

$$d_{i^*j^*} = (c_{i^*j^*} - c'_{i^*j^*})(H^* - H^*)^{-1}.$$

And $SK_{Vi^*} = c_{i^*j^*} - SK_{Vj^*} - H^*d_{i^*j^*}$. As $PK_{Vi^*} = A = aP = SK_{Vi^*}P$, so $\mathcal{B}$ successfully found $a = SK_{Vi^*}$ and solved the ECDLP instance.

Similarly, we can prove that when V_{j^*} or R_{i^*} is not corrupted ($\mathcal{A}$ never made Oracle queries corresponding to V_{j^*} or R_{i^*}), $\mathcal{B}$ can still solve the ECDLP instance by setting $PK_{Vj^*} = A = aP = SK_{Vj^*}P$ or $PK_{Ri^*} = A = aP = SK_{Ri^*}P$. ■

Theorem 2: The proposed APA protocol achieves anonymity assuming the semantic security of the underlying symmetric encryption scheme.

Proof: Suppose there exists a PPT adversary $\mathcal{A}$ which can break the anonymity of the proposed protocol with probability $\epsilon > 1/2$ by outputting a bit $b' = b$ from Test query. We construct a simulator $\mathcal{B}$ to simulate the oracles and utilize $\mathcal{A}$ to break the semantic security of the encryption scheme $SE(\cdot) = (E_k(\cdot), D_k(\cdot))$.

Denote $\Pi_{V_{i^*}}^x$ as the instance that $\mathcal{A}$ queries to test oracle, in particular, V_{i^*} is the target vehicle $\mathcal{A}$ aims to attack for its real identity VID_{i^*} . Let $\mathcal{C}$ be the challenger of CCA game in Definition 2, and $\mathcal{B}$ plays the role of the adversary against it. $\mathcal{C}$ generates K_{TA} for RC, and kept it secret. Then $\mathcal{B}$ generates the other public parameters together with the lists $\mathcal{L}_{Vi}$, $\mathcal{L}_{Ri}$ by following the computations in the system initialization phase, vehicles registration phase, and RSUs registration phase, except for the generation of PID_i ($i \neq i^*$) as below.

- 1) For $i \neq i^*$, $\mathcal{B}$ queries $\langle VID_i, Val_i, r_i \rangle$ to the encryption oracle $E_{K_{TA}}(\cdot)$, and sets PID_i as the oracle output. This is consistent with the adaptive encryption queries made by the adversary in CCA game.
- 2) For $i = i^*$, $\mathcal{B}$ picks two random identities $VID_{i^*}^0$ and $VID_{i^*}^1$, sets $m_0 = \langle VID_{i^*}^0, Val_{i^*}, r_{i^*} \rangle$, $m_1 = \langle VID_{i^*}^1, Val_{i^*}, r_{i^*} \rangle$, sends m_0, m_1 to $\mathcal{C}$. Then $\mathcal{C}$ returns $c^* = E_{K_{TA}}(m_{b^*})$ where $b^* \leftarrow \{0, 1\}$ is a random bit. $\mathcal{B}$ sets $PID_{i^*} = c^*$. This is consistent with the interactions between the adversary and the challenger in Phase 2 of CCA game.

The generation of S_{i^*3} as a random value from $\mathbb{Z}_p$, which is indistinguishable from $r_{i^*} \oplus h(VID_{i^*} \| \sigma_{i^*})$, because $\mathcal{A}$ never queries $VID_{i^*} \| \sigma_{i^*}$ to Hash oracle without knowledge of VID_{i^*} . Therefore, the record for $i = i^*$ in $\mathcal{L}_{Vi}$ should be

$$\{PID_{i^*} = c^*, SK_{Vi^*}, PK_{Vi^*}, Val_{i^*}, Bio_{i^*}, (V_{i^*}, S_{i^*1}, S_{i^*2}, \tau_{i^*})\}.$$

The simulation for the oracles Hash, Execute, Send, Extract, Corrupt are similar as in the proof of Theorem 1. Reveal and Test oracles could be simulated in the following way.

Reveal($\Pi_{V_i}^x$): $\mathcal{B}$ returns VID_i by referring to $\mathcal{L}_{Vi}$ for the record containing V_i , where $i \neq i^*$.

Test($\Pi_{V_{i^*}}^x$): $\mathcal{B}$ returns $VID_{i^*}^0$. Note that in this query, $\mathcal{B}$ does not toss a bit b , however, the operation of returning $VID_{i^*}^0$ actually implies that $b = b^*$. The analysis is given as follows.

- 1) If $b^* = b = 0$: $PID_{i^*} = E_{K_{TA}}(m_0) = E_{K_{TA}}(VID_{i^*}^0, Val_{i^*}, r_{i^*})$, and $\mathcal{B}$ should return $VID_{i^*}^0$ as the real identity of PID_{i^*} in Test query.
- 2) If $b^* = b = 1$: $PID_{i^*} = E_{K_{TA}}(m_1) = E_{K_{TA}}(VID_{i^*}^1, Val_{i^*}, r_{i^*})$, and $\mathcal{B}$ should return a random identity which is not $VID_{i^*}^1$. As $VID_{i^*}^0$ is randomly picked from the identity space and different from $VID_{i^*}^1$, so $\mathcal{B}$ can use $VID_{i^*}^0$ as the answer to Test query.

Finally, if $\mathcal{A}$ successfully outputs a bit $b' = b$ to break the anonymity of APA protocol, then $\mathcal{B}$ uses b' as its output in CCA game. For that $b' = b = b^*$, $\mathcal{B}$ correctly guesses b^* tossed by the challenger $\mathcal{C}$, and thus breaks the semantic security of $SE = (E_k(\cdot), D_k(\cdot))$ for the key $k = K_{TA}$. ■

B. Informal Security Analysis

1) *Stolen-Verifier Attack:* In the proposed protocol, the identity authentication of the vehicle is based on the authentication parameters issued by RC rather than querying the verification table. RC, RSU, and vehicles do not store any validation form. Therefore, our scheme can resist stolen-verifier attack.

2) *Desynchronization Attack:* This attack refers to the adversary interfering with the vehicle's parameters update so that the vehicle cannot complete the subsequent authentication. In the proposed protocol, the vehicle sends a request to the RC to update the pseudo-identity after authentication. Suppose the adversary intercepts $\{PID_i, M_1, T_5\}$ or $\{PID_i, M_2, T_6\}$ to prevent pseudo-identity updating. The vehicle can still use the previous pseudo-identity to complete authentication, signature, and pseudo-identity update.

3) *RSU Captured Attack:* In our protocol, RSU stores $\{RID_i, RSK_{Ri}, PUF(), Cha_i\}$, where $PUF()$ is PUF, which is used in RSU to protect the secret key SK_{Ri} , so even if the adversary launches the RSU captured attack, it will not affect the traffic accident report signed by both parties involved in the accident, and our scheme can resist the RSU captured attacks.

4) *OBU Intrusion Attack:* The vehicle's OBU stores $\{PID_i, VE_i, S_{i1}, S_{i2}, \tau_i\}$, where PID_i is the vehicle's pseudo-identity, $VE_i = h(\sigma_i \| Val_i)$ is the login verification parameter where σ_i is the biological key and Val_i is the vehicle's appearance characteristics, $S_{i1} = SK_{Vi} \oplus h(VE_i \| \sigma_i)$ and $S_{i2} = r_i \oplus h(VID_i \| \sigma_i)$ encrypt the secret number SK_{Vi} and random number r_i , respectively, τ_i is the reproduction parameter. Suppose the adversary obtains the stored message in OBU, he/she cannot recover the secret parameters SK_{Vi} , and r_i without knowing the biological key σ_i . Therefore, the vehicle cannot be impersonated and privacy is protected.

5) *Impersonation Attack:* Suppose an adversary impersonates the vehicle to authenticate or upload a report, he/she

needs to forge $M_{i1} : \{PID_i, AC1, D_{i1}, D_{i2}, c_{i1}, T_1\}$ or $M_2 : \{PID_i, PID_j, AC1, AC2, c_{ij}, D_{ij}, T_1, T_2, VT, T_3\}$. In M_{i1} , where $c_{i1} = SK_{Vi} + h(AC1||D_{i1}||D_{i2}||T_1||PID_i||Val_i) * d_{i1}$, SK_{Vi} and σ'_i are vehicle's secret key and the user's biological key, respectively, they are unavailable. Similarly, in M_2 , $VT = h(SK_{Vi} \cdot PK_{Rt}||T_3)$ is unforged and cannot be replayed. Therefore, the vehicle cannot be impersonated.

6) *Replay Attack*: In our protocol, timestamps and random numbers are combined in messages to resist replay attacks. The replayed messages cannot pass the verification.

7) *Anonymity and Unlinkability*: $PID_i = E_{K_{TA}}(VID_i, Val_i, r_i)$ is vehicle's pseudo-identity, where K_{TA} is RC's secret key, r_i is a random number. The updated pseudo-identity $PID_i^* = E_{K_{TA}}(VID_i||Val_i||r_i^*)$ where r_i^* is independent of r_i . Therefore, the pseudo-identities are anonymous and unlinkable.

8) *Malicious Message Tracking*: Suppose a legitimate user sends malicious messages $M_{i1} : \{PID_i, AC1, D_{i1}, D_{i2}, c_{i1}, T_1\}$ or $M_2 : \{PID_i, PID_j, AC1, AC2, c_{ij}, D_{ij}, T_1, T_2, VT, T_3\}$, RC can recover the real identity VID_i by calculating $(VID_i, Val_i, r_i) = D_{K_{TA}}(PID_i)$ to track malicious messages.

9) *Two-Factor Secrecy*: According to the definition of two-factor authentication, an adversary can obtain either driver's biological information or OBU storage, but not both.

Suppose an adversary can obtain the driver's biological information Bio_i , it is infeasible to compute $\sigma'_i = Rep(Bio'_i, \tau_i)$, $VE_i = h(\sigma'_i||Val_i)$, and $SK_{Vi} = S_{i1} \oplus h(VE_i||\sigma'_i)$ without the knowledge of τ_i , Val_i and S_{i1} stored in OBU. On the other hand, suppose an adversary can access $\{PID_i, VID_i, VE_i, S_{i1}, S_{i2}, \tau_i\}$ stored in the vehicle's OBU, it is infeasible to compute SK_{Vi} without the knowledge of the driver's biological information Bio_i .

Conclusively, our scheme achieves two-factor authentication.

VI. PERFORMANCE ANALYSIS AND COMPARISON

In this section, we compare the proposed scheme with some related works [5], [10], [12], [16], [17] in terms of security and efficiency, including security properties and robustness against various attacks, blockchain platform setting and specifics, performance analysis on computational costs, and communication overhead and latency.

A. Security Comparison

Table III presents comparison results of representative VANETs protocols in this research filed, regarding many desirable security properties and robustness against various attacks. Conclusively, our protocol is the only one among them that properly addresses all the security issues. Detailed analysis about some security weaknesses of the other protocols in Table III is given below.

In [5], the protocol suffers from collusion attack of witness vehicles which are invited to jointly confirm the traffic accident report. Besides, it cannot resist replay attack for that the receiver can neither identify the replayed messages nor verify the real identity of the sender. In [10], the protocol does not support identity anonymity and unlinkability because vehicles use static pseudonyms, and it is vulnerable to OBU intrusion

TABLE III
COMPARISON OF SECURITY PROPERTIES

Attacks/Properties	[5]	[10]	[12]	[16]	[17]	Ours
Privileged-Insider Attack	x	✓	✓	✓	✓	✓
Off-line Password Guessing Attack	-	-	-	-	-	✓
Impersonation Attack	x	x	x	✓	x	✓
Replay Attack	x	✓	x	✓	✓	✓
Man-in-Middle Attack	✓	✓	✓	✓	✓	✓
OBUs Intrusion Attack	x	x	x	x	x	✓
RSU Captured Attack	✓	x	x	x	x	✓
Stolen-Verifier Attack	x	✓	✓	✓	✓	✓
Update asynchronous Attack	-	-	-	✓	-	✓
Identity Anonymity	x	x	x	✓	✓	✓
Mutual Authentication	x	✓	x	✓	x	✓
Batch Verification	x	✓	✓	x	✓	✓
Malicious Message Tracking	✓	✓	x	x	✓	✓
Unlinkability	x	x	x	✓	x	✓

✓:Resist(Attacks)/Possess(Properties) x:Suffer(Attacks)/No(Properties)
-: N/A

attacks and RSU captured attacks. In [12], replay attack may be applied to interfere with the vehicles' communications, while OBU intrusion attack and impersonation attack could be utilized to corrupt the security since OBU stores key pair in plaintext. Furthermore, it fails to provide resistance against RSU captured attack and malicious message tracing attack. In [16], it also suffers from RSU captured attack since an attacker can authorize illegal users by capturing RSUs. In [17], the protocol is found insecure against OBU intrusion attack and impersonation attack, while vehicle's identities are linkable, due to similar causes as in [10] and [12] mentioned above.

B. Performance Comparison

We simulated the overhead of the proposed scheme in the environment of Raspberry Pi 4B quad-core 64-bits ARM Cortex-A72, 1.5 GHz, 2-GB LPDDR4 SDARM, and evaluated the performance of vehicles and RSUs when processing multiple requests. Raspberry Pi is a lightweight microcomputer with extraordinary scalability and development potential, which offers compatibility with various operating systems and supports expansion to multiple wireless communication modes. It has been widely used in VANETs as the control core of vehicles. Regarding the setting of blockchain platform, it is based on Hyperledger Fabric [26] and integrated with VANETs using the gRPC framework, which adds an extra security layer to the network for ensuring that only authorized nodes are aware of vehicles' identities. Hyperledger Fabric offers modular consensus options, including Kafka-based service, which can be fine-tuned for shorter transaction confirmation times. Since large block size may result in longer propagation times and increase the risk of network congestion in VANET environments, we adopt dynamic block size adjustment mechanism according to network conditions [27], by setting a threshold of block sizes which can be adjusted dynamically to accommodate fluctuations in transaction volume for preventing network congestion.

In Table IV, we analyze the overhead of our protocol in the signature and verification phases by comparison with related schemes. We use T_{exp} , T_h , T_{mul} , T_{ecc} , T_{bp} , T_{ead} , T_{ECE_SV} , T_{HE_SV} , T_{HE_ED} , and T_{SED} to represent the running time required for one exponentiation, one hash, one

TABLE IV
COMPARISON OF COMPUTATIONAL COSTS

Scheme	Signature (single vehicle)		Verification	Batch Verification (n vehicles)	Total (sign+verify)	Time (ms)
	Vehicle	RSU				
[5]	$i(3T_{ECC_{SV}} + T_{ecc} + T_{SED} + T_h)$	-	$T_{ECC_{SV}} + T_{SED} + T_h$	$n(T_{ECC_{SV}} + T_{SED} + T_h)$	$(3i+1)T_{ECC_{SV}} + iT_{ecc} + (i+1)(T_{SED} + T_h)$	35.164ms ($i=1$)
[10]	$3T_{ecc} + 3T_h + T_{ead} + T_{HE_{ED}}$	$T_{ecc} + T_h$	$5T_{ecc} + 2T_h + 2T_{ead} + T_{HE_{ED}}$	$(2n+3)T_{ecc} + 2nT_h + 2T_{ead} + T_{HE_{ED}}$	$9T_{ecc} + 6T_h + 3T_{ead} + 2T_{HE_{ED}}$	44.00ms
[12]	$3T_{exp} + 2T_{mul} + T_h$	$T_{exp} + T_{mul}$	$2T_{mul} + T_{exp}$	$nT_{exp} + (2n-1)T_{mul}$	$4T_{exp} + 4T_{mul} + T_h$	20.215ms
[16]	$3T_{HE_{ED}} + T_{mul}$	$T_{exp} + 2T_{mul} + T_{HE_{ED}} + T_{HE_{SV}}$	-	$8nT_{exp} + 3nT_{mul} + 3nT_{HE_{ED}}$	$9T_{exp} + 6T_{mul} + 7T_{HE_{ED}} + T_{HE_{SV}}$	127.607ms
	V2V: $8T_{exp} + 3T_{mul} + 3T_{HE_{ED}}$					
[17]	$T_h + T_{mul} + T_{exp} + T_{ecc}$	$2T_h + T_{bp} + T_{ecc}$	$T_h + T_{ecc} + T_{bp} + 2T_{ead}$	$nT_h + nT_{ecc} + T_{bp} + 2nT_{ead}$	$4T_h + T_{mul} + 3T_{ecc} + 2T_{bp} + T_{exp} + 2T_{ead}$	102.023ms
ours	$4T_h + 4T_{ecc} + 2T_{ead} + T_{mul}$	-	$4T_h + 4T_{ecc} + 2T_{ead} + T_{mul}$	$(5+3n)T_h + (n+1)T_{mul} + (2n+2)T_{ead} + (7+n)T_{ecc}$	$8T_h + 8T_{ecc} + 4T_{ead} + 2T_{mul}$	21.146ms

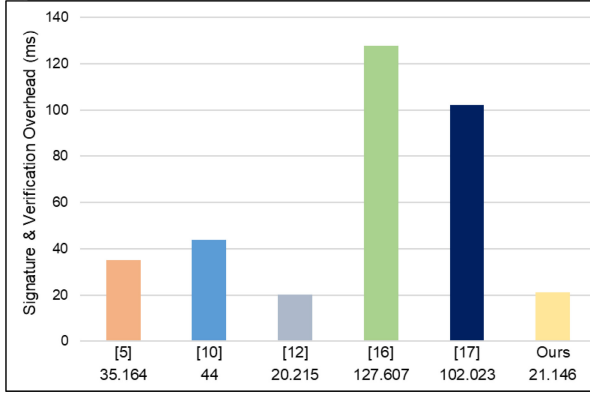


Fig. 6. Overhead of single signature and verification.

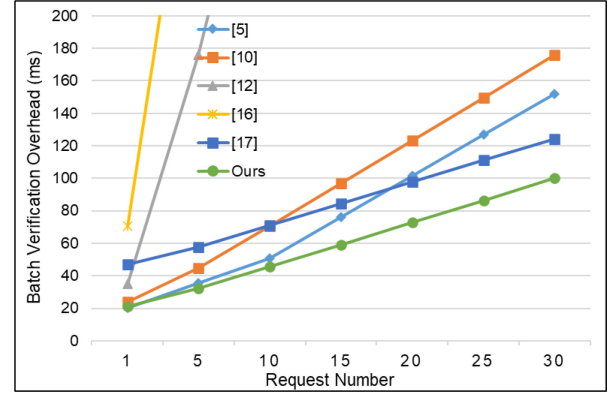


Fig. 7. Overhead of batch verification.

multiplication, one ECC point multiplication, one bilinear pairing, one ECC point addition, one EC-Elgamal signature/verification, one Paillier signature/verification, one Paillier encryption/decryption, one symmetric encryption/decryption, respectively. According to the simulation results in the above environment, $T_{exp} \approx 5.026$ ms, $T_h \approx 0.019$ ms, $T_{mul} \approx 0.023$ ms, $T_{ecc} \approx 2.610$ ms, $T_{bp} \approx 44.517$ ms, $T_{ead} \approx 0.017$ ms, $T_{ECC_{SV}} \approx 7.849$ ms, $T_{HE_{SV}} \approx 11.024$ ms, $T_{HE_{ED}} \approx 10.173$ ms, and $T_{SED} \approx 0.56$ ms.

As shown by the statistics in Table IV, our protocol enjoys the lowest level of computational overhead except for Chen et al.'s [12] scheme regarding single signature and verification (see Fig. 6 for clarity), simultaneously achieves the minimal batch verification cost among the related schemes (see Fig. 7). Furthermore, our protocol eliminates the cost of RSU besides Liu et al.'s [5] scheme in the signature generation phase for accident reports.

In Table V, we present the comparative results of communication overhead and latency. Without loss of generality, we assume the candidates of hash function and symmetric encryption are SHA-256 and AES-128, respectively, the sizes of identity, password, or timestamp are 32bits, and the sizes of other random nonces are 256 bits. The communication cost and latency of related schemes are affected by the number of participating vehicles and the message delay. As we can see, our scheme outperforms others in terms of communicational efficiency, even in the ideal situation that vehicles respond messages simultaneously.

TABLE V
COMPARISON OF COMMUNICATION OVERHEAD AND LATENCY

Schemes	V2V		V2R	
	Cost	Latency	Cost	Latency
[5]	2208L bits	>27 ms	869M bits	>36 ms
[10]	-	-	1376M+832 bits	>44 ms
[12]	512W bits	>31 ms	512V+512W+1024 bits	>21 ms
[16]	2976 bits	71 ms	832 bits	128 ms
[17]	-	-	1152+577Mbits	>102 ms
ours	2240 bits	16 ms	704 bits	22 ms

*L denotes the cardinality of ID set. M denotes the number of received messages. V denotes the number of surrounding vehicles. W denotes the number of selected witness vehicles.

Regarding the dynamic pseudo-identity strategy applied in our protocol, it causes the overhead of temporary identity update (see Section IV-E). The operation of updating one pseudo-identity involves one-round communication between the vehicle and RC, in which the vehicle challenges RC with a new identity and then stores the updated parameters in OBU after receiving the response from RC. The computational cost is $4T_h + T_{ecc} + 2T_{SED} \approx 3.806$ ms on the side of vehicle and $4T_{SED} \approx 2.24$ ms on the side of RC, respectively, while the communication overhead is 192 bits on either side. As we can see, the additional costs of dynamic identities are almost negligible compared to overall overhead. On the other hand, static identities require for extra storage on the server to keep them in a table for comparison in the authentication. From the perspective of preserving identity privacy, dynamic identities are also more preferable than static identities, since the

security still holds even in the scenarios of server compromise and channel exposure.

Network partitioning and latency are common in real-world vehicle networks due to vehicle mobility, signal interference, and different network conditions. In the proposed scheme, RSU nodes jointly maintain the blockchain to get rid of the issue of network partitioning. Furthermore, the latency of our protocol is not affected by the number of vehicles and transmitted messages, hence the commonly adopted asynchronous communication methods, timeout and retry mechanisms in network communication can be normally deployed in our protocol without compatibility problems.

VII. CONCLUSION

In this article, we found that existing traffic accident handling schemes cannot be applied to the scenarios without RSUs, and are prone to cause disputes or suffer from RSU captured attacks and collusion attacks of witness vehicles. To address the above issues, a blockchain-based V2V and V2R authentication and traffic accident handling protocol is proposed. Especially, accident reports can be generated and confirmed solely by involved vehicles and realize nonrepudiation and public verifiability. Formal proof and analysis are conducted to validate the correctness and security of the proposed protocol. By simulation experiments, we compare our protocol with related schemes and conclude that ours is more secure while maintaining a low level of overhead.

Due to the support of batch verification, the proposed scheme applies to the handling of accident reports in the scenario of concurrent traffic accidents. Besides, with the advantage of blockchain utilization, our scheme supports relevant authorities (e.g., traffic police, and insurance companies) to investigate, audit, and collect evidence effectively after the accidents. For future work, the potential of blockchain could be further explored by designing and deploying appropriate smart contracts to collaborate with other entities in our model, in order to realize automatization and higher efficiency of the traffic accident handling procedure.

REFERENCES

- [1] D. He, S. Zeadally, B. Xu, and X. Huang, "An efficient identity-based conditional privacy-preserving authentication scheme for vehicular ad hoc networks," *IEEE Trans. Inf. Forensics Security*, vol. 10, pp. 2681–2691, 2015.
- [2] J. C. Mukherjee, A. Gupta, and R. C. Sreenivas, "Event notification in VANET with capacitated roadside units," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, pp. 1867–1879, Jul. 2016, doi: [10.1109/TITS.2015.2511145](#).
- [3] M. Azees, P. Vijayakumar, and L. J. Deboarh, "EAAP: Efficient anonymous authentication with conditional privacy-preserving scheme for vehicular ad hoc networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 9, pp. 2467–2476, Sep. 2017, doi: [10.1109/TITS.2016.2634623](#).
- [4] J. Cui, J. Zhang, H. Zhong, and Y. Xu, "SPACF: A secure privacy-preserving authentication scheme for VANET with cuckoo filter," *IEEE Trans. Veh. Technol.*, vol. 66, no. 11, pp. 10283–10295, Nov. 2017.
- [5] X. Liu, H. Huang, F. Xiao, and Z. Ma, "A blockchain-based trust management with conditional privacy-preserving announcement scheme for VANETs," *IEEE Internet Things J.*, vol. 7, no. 5, pp. 4101–4112, May 2020, doi: [10.1109/JIOT.2019.2957421](#).
- [6] C.-L. Chen, Y.-X. Chen, C.-F. Lee, Y.-Y. Deng, and C.-H. Chen, "An efficient and secure key agreement protocol for sharing emergency events in VANET systems," *IEEE Access*, vol. 7, pp. 148472–148484, 2019, doi: [10.1109/ACCESS.2019.2946969](#).
- [7] J. Zhang, J. Cui, H. Zhong, Z. Chen, and L. Liu, "PA-CRT: Chinese remainder theorem based conditional privacy-preserving authentication scheme in vehicular ad-hoc networks," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 2, pp. 722–735, Mar./Apr. 2021, doi: [10.1109/TDSC.2019.2904274](#).
- [8] H. Xiong, J. Chen, Q. Mei, and Y. Zhao, "Conditional privacy-preserving authentication protocol with dynamic membership updating for VANETs," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 3, pp. 2089–2104, May/Jun. 2022, doi: [10.1109/TDSC.2020.3047872](#).
- [9] Q. Xie and J. Huang, "Improvement of a conditional privacy-preserving and desynchronization-resistant authentication protocol for IoV," *Appl. Sci.-Basel*, vol. 14, no. 6, p. 2451, 2024, doi: [10.3390/app14062451](#).
- [10] H. Cheng, M. Shojafar, M. Alazab, R. Tafazolli, and Y. Liu, "PPVF: Privacy-preserving protocol for vehicle feedback in cloud-assisted VANET," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 9391–9403, Jul. 2022, doi: [10.1109/TITS.2021.3117950](#).
- [11] Q. Mei, H. Xiong, Y.-C. Chen, and C.-M. Chen, "Blockchain-enabled privacy-preserving authentication mechanism for transportation CPS with cloud-edge computing," *IEEE Trans. Eng. Manag.*, early access, Apr. 14, 2022, doi: [10.1109/TEM.2022.3159311](#).
- [12] X. Chen, A. Yang, Y. Tong, J. Weng, J. Weng, and T. Li, "A multisignature-based secure and OBU-friendly emergency reporting scheme in VANET," *IEEE Internet Things J.*, vol. 9, no. 22, pp. 23130–23141, Nov. 2022, doi: [10.1109/JIOT.2022.3184991](#).
- [13] M. A. R. Bae, L. Simpson, X. Boyen, E. Foo, and J. Pieprzyk, "ALI: Anonymous lightweight inter-vehicle broadcast authentication with encryption," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 3, pp. 1799–1817, May/Jun. 2022, doi: [10.1109/TDSC.2022.3164436](#).
- [14] Q. Xie, Z. Ding, and P. Zheng, "Provably secure and anonymous V2I and V2V authentication protocol for VANETs," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7318–7327, Jul. 2023, doi: [10.1109/TITS.2023.3253710](#).
- [15] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," in *Proc. Decentralized Bus. Rev.*, 2008, Art. no. 21260. [Online]. Available: <https://bitcoin.org/bitcoin.pdf>
- [16] X. Li, J. Liu, M. S. Obaidat, P. Vijayakumar, Q. Jiang, and R. Amin, "An unlinkable authenticated key agreement with collusion resistant for VANETs," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 7992–8006, Aug. 2021, doi: [10.1109/TVT.2021.3087557](#).
- [17] L. Zhang and J. Xu, "Blockchain-based anonymous authentication for traffic reporting in VANETs," *Connect. Sci.*, vol. 34, no. 1, pp. 1038–1065, 2022, doi: [10.1080/09540091.2022.2026888](#).
- [18] Q. Xie, Z. Ding, W. Tang, D. He, and X. Tan, "Provable secure and lightweight blockchain-based V2I handover authentication and V2V broadcast protocol for VANETs," *IEEE Trans. Veh. Technol.*, vol. 72, no. 12, pp. 15200–15212, Dec. 2023, doi: [10.1109/TVT.2023.3289175](#).
- [19] K. N. Qureshi, G. Jeon, M. M. Hassan, M. R. Hassan, and K. Kaur, "Blockchain-based privacy-preserving authentication model intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7435–7443, Jul. 2023, doi: [10.1109/TITS.2022.3158320](#).
- [20] J. Zhao, F. Huang, L. Liao, and Q. Zhang, "Blockchain-based trust management model for vehicular ad hoc networks," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 8118–8132, Mar. 2024, doi: [10.1109/JIOT.2023.3318597](#).
- [21] S. Roy, S. Nandi, R. Maheshwari, S. Shetty, A. K. Das, and P. Lorenz, "Blockchain-based efficient access control with handover policy in IoV-enabled intelligent transportation system," *IEEE Trans. Veh. Technol.*, vol. 73, no. 3, pp. 3009–3024, Mar. 2024, doi: [10.1109/TVT.2023.3322637](#).
- [22] G. Du et al., "A blockchain-based trust-value management approach for secure information sharing in Internet of Vehicles," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 333–344, Jan. 2024, doi: [10.1109/JIOT.2023.3277691](#).
- [23] L. Wei, J. Cui, H. Zhong, I. Bolodurina, and L. Liu, "A lightweight and conditional privacy-preserving authenticated key agreement scheme with multi-TA model for fog-based VANETs," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 1, pp. 422–436, Jan./Feb. 2023, doi: [10.1109/TDSC.2021.3135016](#).
- [24] Q. Xie, Z. Sun, Q. Xie, and Z. Ding, "A cross-trusted authority authentication protocol for Internet of Vehicles based on blockchain," *IEEE Access*, vol. 11, pp. 97840–97851, 2023, doi: [10.1109/ACCESS.2023.3308601](#).
- [25] A. Alharthi, Q. Ni, R. Jiang, and M. A. Khan, "A computational model for reputation and ensemble-based learning model for prediction of trustworthiness in vehicular ad hoc network," *IEEE Internet Things J.*, vol. 10, no. 20, pp. 18248–18258, Oct. 2023, doi: [10.1109/JIOT.2023.3279950](#).

- [26] E. Androulaki et al., "Hyperledger fabric: A distributed operating system for permissioned blockchains," in *Proc. 13th EuroSys Conf.*, 2018, pp. 1–15, doi: [10.1145/3190508.3190538](https://doi.org/10.1145/3190508.3190538).
- [27] J. Wang et al., "BPR: Blockchain-enabled efficient and secure parking reservation framework with block size dynamic adjustment method," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3555–3570, Mar. 2023, doi: [10.1109/TITS.2022.3222960](https://doi.org/10.1109/TITS.2022.3222960).



Qi Xie received the Ph.D. degree in applied mathematics from Zhejiang University, Hangzhou, China, in 2005.

He is a Professor with Hangzhou Normal University, Hangzhou, and a Director of the Key Laboratory of Cryptography of Zhejiang Province, Hangzhou Normal University. He was a Visiting Scholar from 2009 to 2010 with the Department of Computer Science, University of Birmingham, Birmingham, U.K., and also with the Department of Computer Science, The City University of Hong

Kong, Hong Kong, in 2012. He has published over 100 research papers in international journals and conferences, such as *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY*, *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, and *IEEE INTERNET OF THINGS JOURNAL*. His research area is applied cryptography, including digital signatures, authentication, and key agreement protocols.

Prof. Xie served as a General Co-Chair for ISPEC2012 and ACM ASIACCS2013, and a reviewer for over 40 international journals.



Zixuan Ding received the master's degree from Hangzhou Normal University, Hangzhou, China, in 2023. He is currently pursuing the Doctoral degree with Nankai University, Tianjin, China. He mainly studies authentication protocols and cryptography.



Qingyun Xie is currently pursuing the bachelor's degree with the School of Information Science and Technology, Hangzhou Normal University, Hangzhou, China.

Her research interest is mainly in authentication protocols.



Xiao Tan received the B.S. and M.S. degrees from Fudan University, Shanghai, China, in 2007 and 2010, respectively, and the Ph.D. degree from The City University of Hong Kong, Hong Kong, in 2013.

He is a Lecturer with Hangzhou Normal University, Hangzhou, China, and is a Researcher with the Key Laboratory of Cryptography of Zhejiang Province, Hangzhou Normal University. His main research interests include cryptography and information security, in particular, digital signatures,

authenticated key agreement, and encryption schemes.



Debiao He (Member, IEEE) received the Ph.D. degree in applied mathematics from the School of Mathematics and Statistics, Wuhan University, Wuhan, China, in 2009.

He is currently a Professor with the School of Cyber Science and Engineering, Wuhan University. He has published over 100 research papers in refereed international journals and conferences, such as *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY* and *Usenix Security Symposium*. His work has been cited more than

10 000 times at Google Scholar. His main research interests include cryptography and information security, in particular, cryptographic protocols.

Prof. He is the recipient of the 2018 IEEE Systems Journal Best Paper Award and the 2019 IET Information Security Best Paper Award. He is in the editorial board of several international journals, such as *Journal of Information Security and Applications*, *Frontiers of Computer Science*, and *Human-Centric Computing and Information Sciences*.



Wen Tang is currently pursuing the M.S. degree with the School of Information Science and Technology, Hangzhou Normal University, Hangzhou, China. She mainly studies authentication and key agreement protocols.